ASSIGNMENT REPORT- 7

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For The Subject

Pattern Recognition and Machine Learning

(26AI5707)



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INTRODUCTION:

This assignment implements Gaussian Mixture Models for pattern recognition. In Part 1 two synthetic classes are generated from mixtures of three two- Gaussian components. Separate Gaussian Mixture Models are trained for the two classes using eighty percent of the generated samples and classification is evaluated using confusion matrices under diagonal and full covariance. In Part 2 Gaussian Mixture Models are trained for three MNIST digits—1, 6 and 9—using the training images only. Each 28 by 28 grayscale image is represented by seven hundred eighty-four pixel features and a diagonal-covariance Gaussian Mixture Model is trained for each digit. The GMM classify the test images by comparing class log-likelihood scores.

METHODOLOGY:

1. Random number and Gaussian data generation

A deterministic random-number generator is implemented using a Linear Congruential Generator (LCG). The gaussian() function uses the Marsaglia polar method to generate standard normal random values. The generate_gaussian() function transforms two independent standard normal values to have the requested two-dimensional mean and covariance using a Cholesky-style transformation.

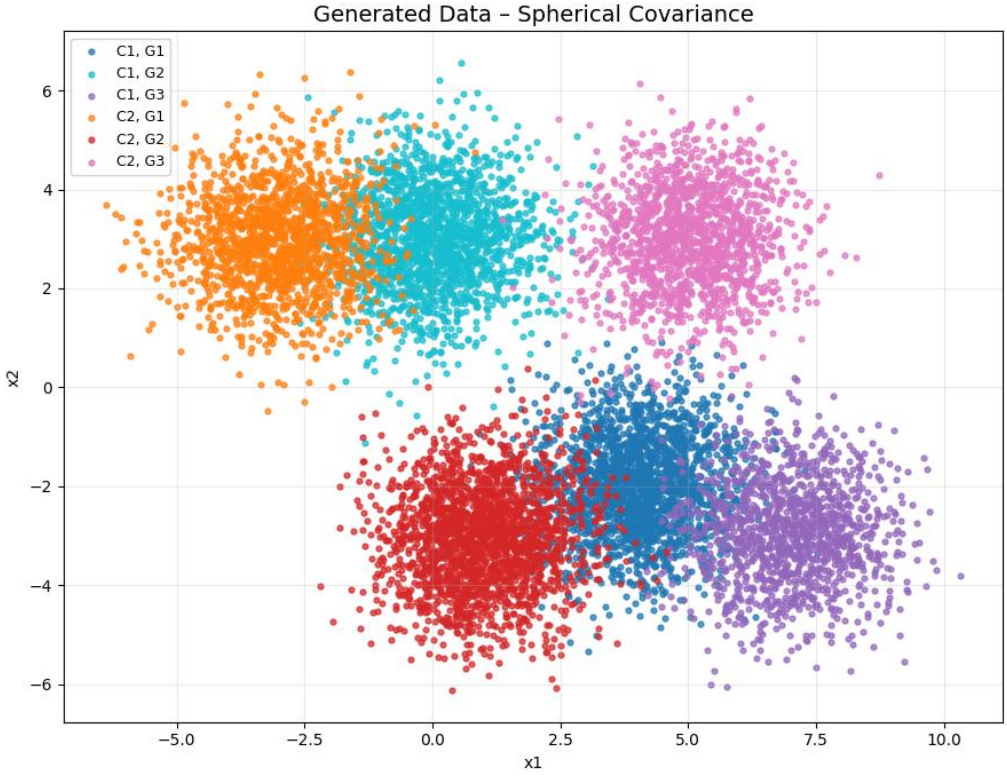
$$x_{n+1} = (a\, x_n + c) \bmod m$$

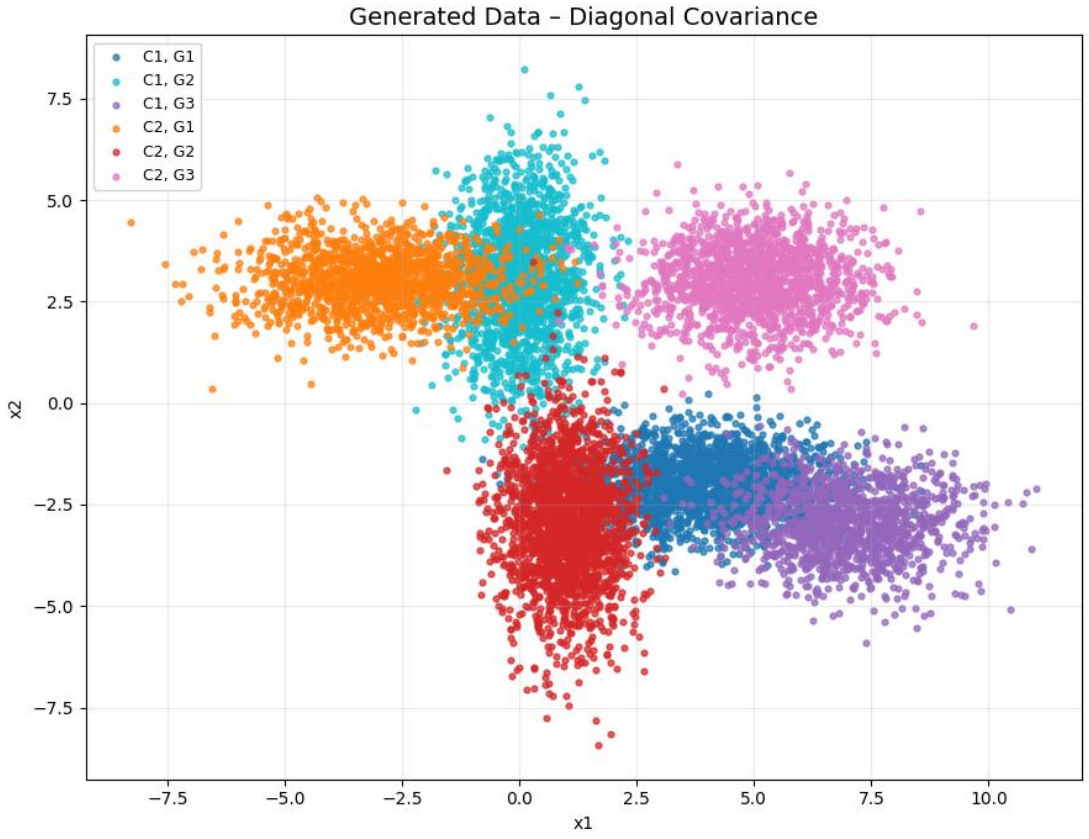
where,

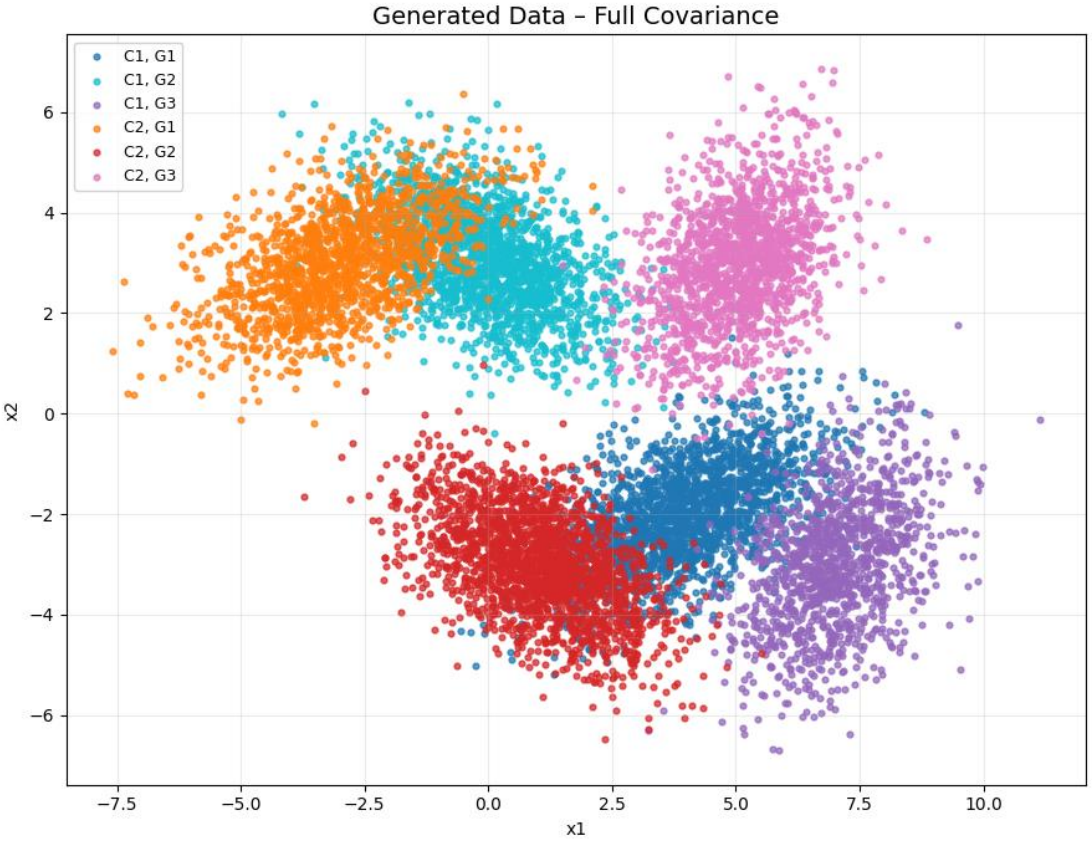
a = 1664525

c = 1013904223

$m = 2^{32}$




Generated Data – Diagonal Covariance
C1, G1
C1, G2
C1, G3
C2, G1
C2, G2
C2, G3
x1
x2


Generated Data – Full Covariance
C1, G1
C1, G2
C1, G3
C2, G1
C2, G2
C2, G3
x1
x2

2. Gaussian Mixture Models (GMM) and Expectation-Maximization (EM)

The Gaussian log-probability function evaluates the multivariate Gaussian density in log form. The implementation uses the covariance log-determinant and a linear-system solution to calculate the quadratic Mahalanobis-distance term. The logsumexp() function combines component log probabilities in a numerically stable way.

The EM algorithm estimates the GMM parameters - Initialization (component means are selected from training observations; covariance matrices are initialized from the global covariance; mixture weights start equally), E-step (calculate each observation's responsibility for each component, representing the posterior membership probability), M-step (update component weights, means and covariance matrices using the responsibilities), Convergence(compare the current average log-likelihood with the previous value and stop when the absolute difference is below the tolerance, or when the iteration limit is reached).

3. Synthetic-data

Covariance model	Confusion matrix	Accuracy
Spherical	[[952, 48], [57, 943]]	94.75%
Diagonal	[[942, 58], [44, 956]]	94.90%
Full	[[909, 91], [87, 913]]	91.10%

4. EM convergence

The average training log-likelihood across EM iterations for the two class models. The curves generally rise and then flatten, indicating that parameter updates are approaching a stable solution.

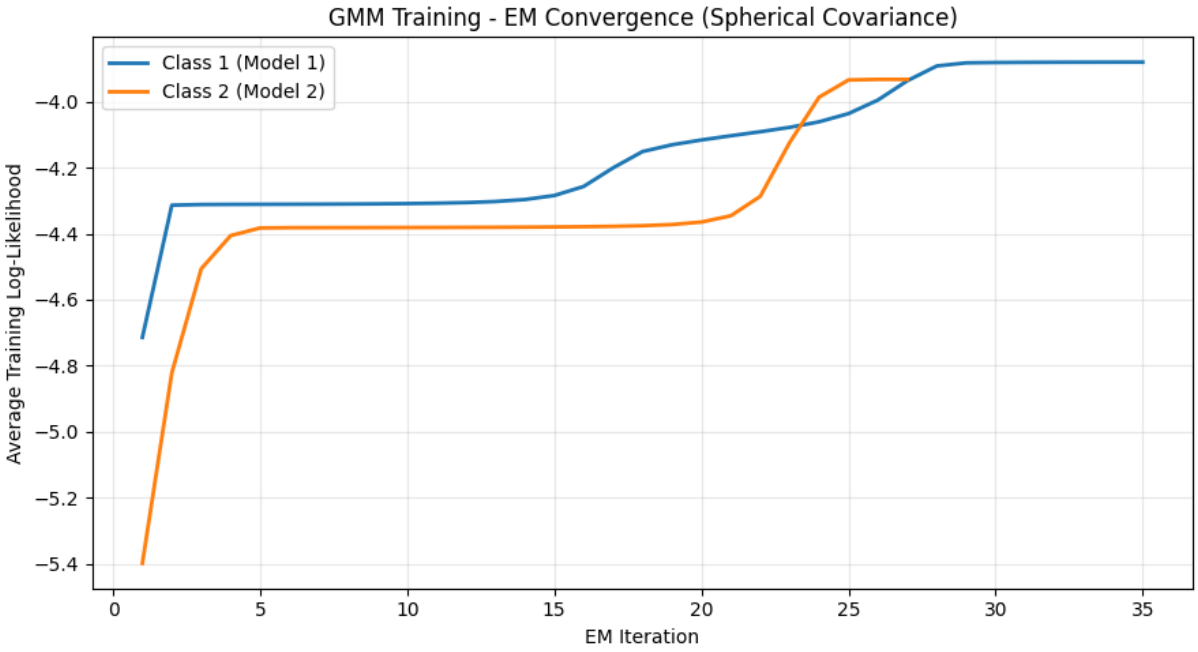
$$| L_{\text{current}} - L_{\text{previous}} | < \epsilon$$

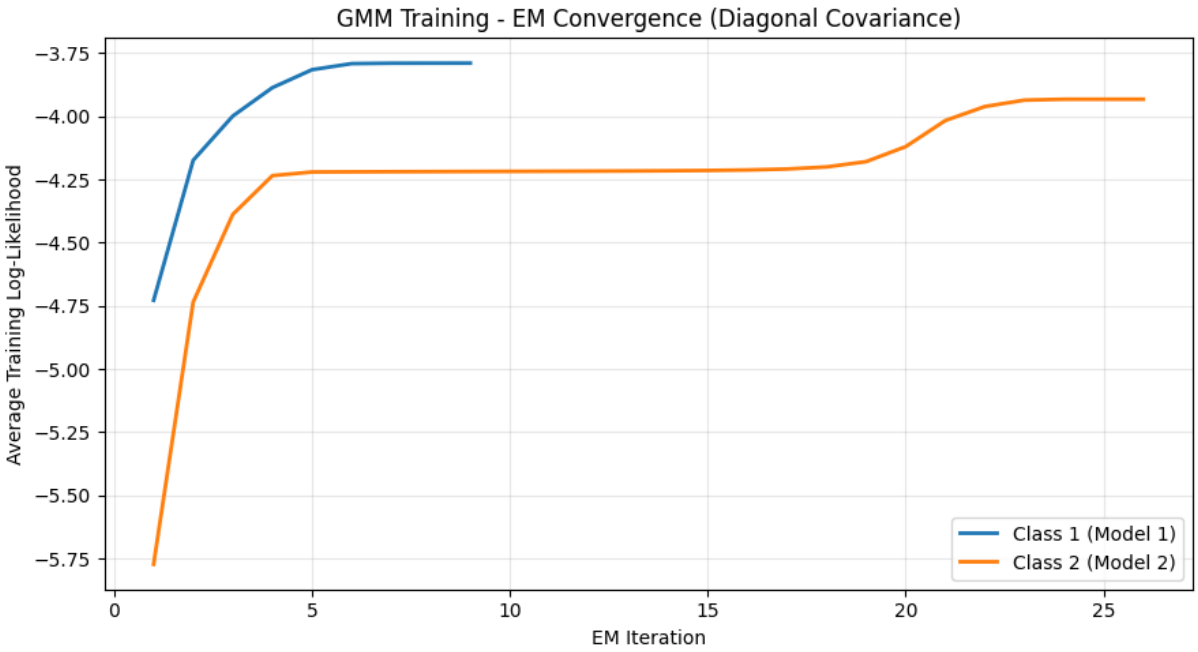
Where:

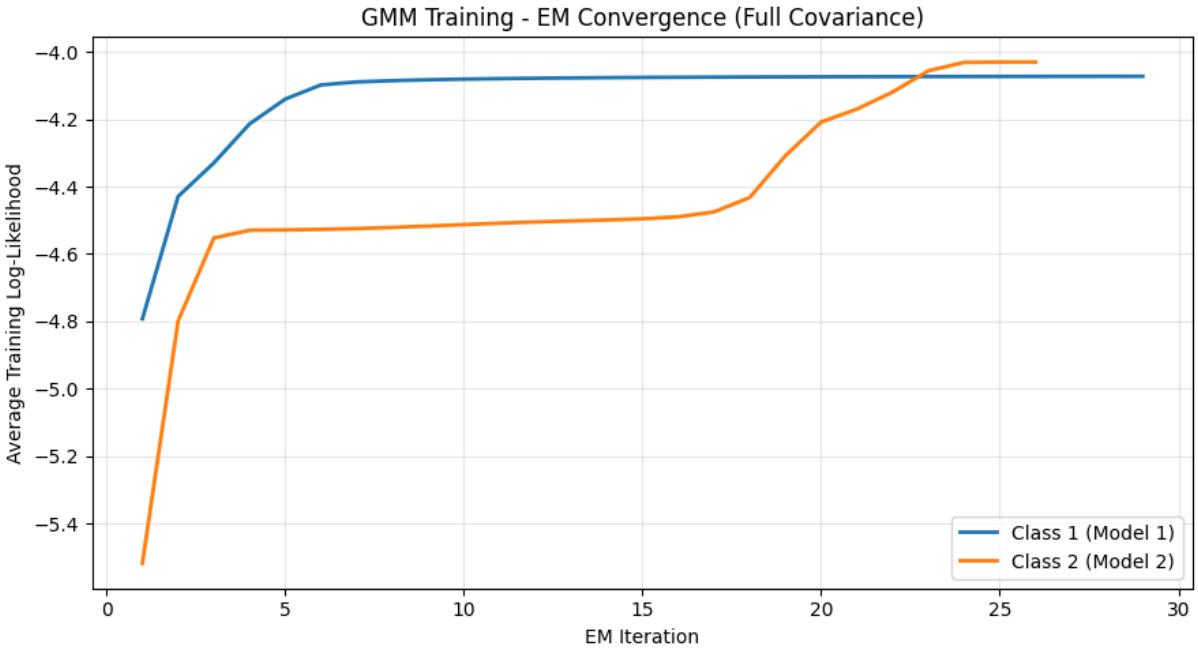
L_{current} = Current iteration's average log-likelihood.

L_{previous} = Previous iteration's average log-likelihood.

ϵ = Convergence tolerance is 10^{-4}







5. Training and Testing Split

The samples are randomly permuted and divided into 80% training data and 20% test data separately for each class. Therefore, each class contributes 4,000 training points and 1,000 test points for each covariance.

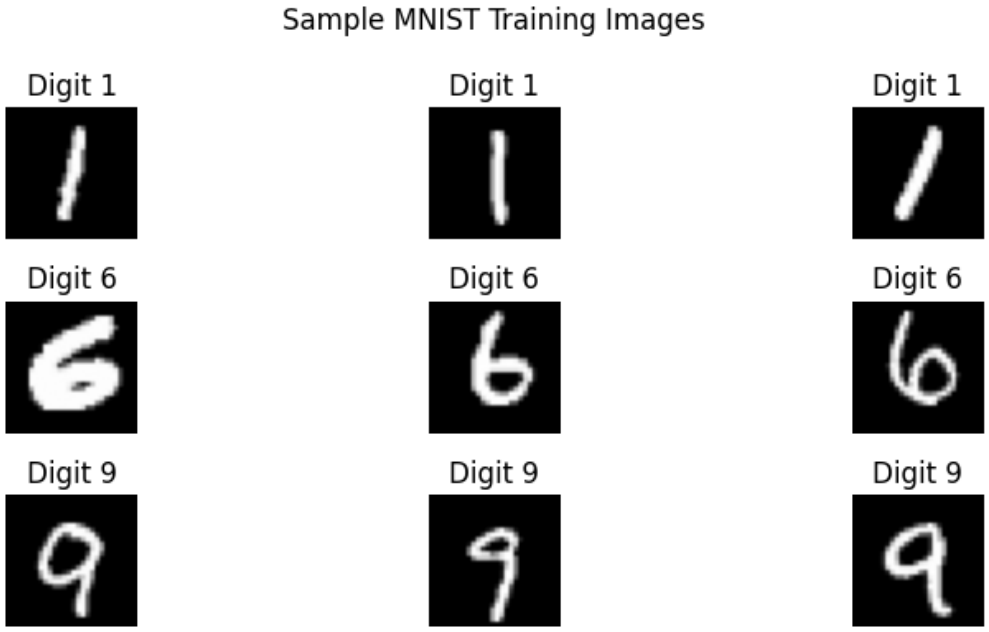
6. Dataset and selected classes

The provided dataset contains train, dev and test folders for digits 1, 6 and 9. The code loads the image folders. The code loaded their respective images for GMM training and testing.

Digit	**Training images**	**Test images**
1	5,383	1,135
6	4,752	958
9	4,776	1,009

7. Image preprocessing

The images are converted to grayscale, resized to 28 × 28 pixels and converted to a floating-point NumPy array. Pixel values are normalized by dividing by 255, mapping the grayscale intensity range to [0, 1]. A 28 × 28 image contains 784 pixel values and is represented as a feature vector for GMM modelling.

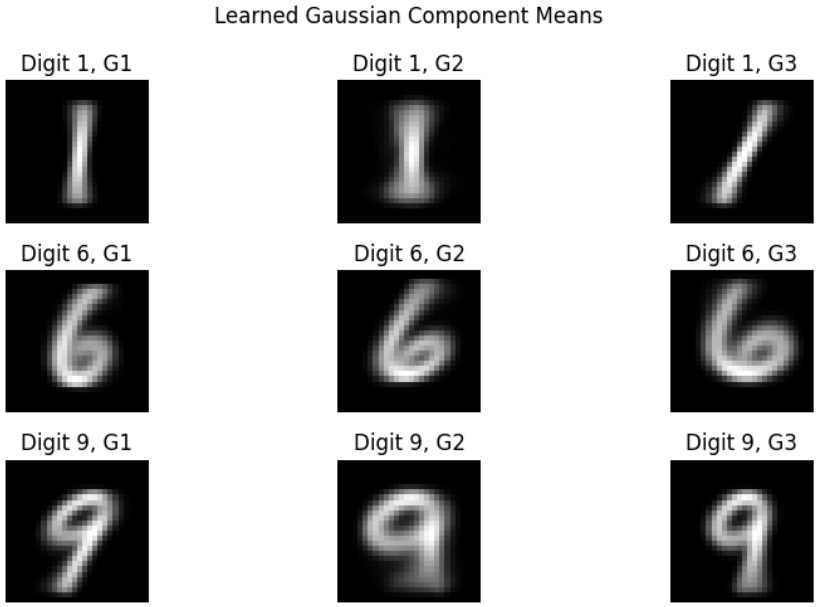


8. Diagonal-covariance GMM

A separate DiagonalGMM model is trained for each digit. Each model uses K = 3 Gaussian components, up to 100 EM iterations, tolerance 10^{-4} and variance regularization 10^{-4}. The component log-probability is calculated from the mixture weight, per-feature variance and squared difference between an image vector and the component mean.

9. Learned Gaussian component means

After training, each component mean contains 784 pixel-intensity values and is reshaped to 28 × 28 for visualization. They illustrate the visual patterns represented by the three Gaussian components within each digit class.



Digit order: [1, 6, 9]

Confusion Matrix
Rows = Actual digits
Columns = Predicted digits
[[1087 21 27]
 [7 947 4]
 [16 4 989]]

Overall Test Accuracy: 97.45%
CLASSIFICATION PERFORMANCE
Digit 1
Correct predictions: 1087
Total test images: 1135
Recall: 95.77%

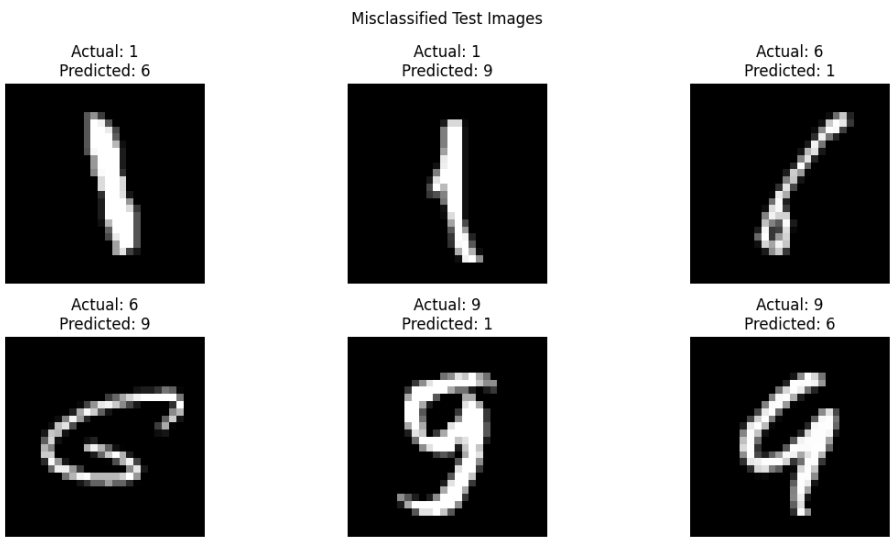
Digit 6
Correct predictions: 947
Total test images: 958
Recall: 98.85%

Digit 9
Correct predictions: 989
Total test images: 1009
Recall: 98.02%

10. Misclassification and class-boundary analysis

A handwritten digit can vary in slant, stroke thickness, curvature and loop closure. These visual variations can make an image resemble another class. The classifier selects the model with the largest likelihood, so an image near an overlap region between class distributions can be assigned to the wrong digit.

Actual 1 -> Predicted 6: 21 errors
Actual 1 -> Predicted 9: 27 errors
Actual 6 -> Predicted 1: 7 errors
Actual 6 -> Predicted 9: 4 errors
Actual 9 -> Predicted 1: 16 errors
Actual 9 -> Predicted 6: 4 errors



LIKELIHOOD MARGIN ANALYSIS
Actual digit: 1
Predicted digit: 6
Log-likelihood scores: [1610.77688152 1620.72301322 1131.75705964]
Likelihood margin: 9.946131691439405

Actual digit: 1
Predicted digit: 9
Log-likelihood scores: [1597.48155674 1166.68864954 1758.31842201]
Likelihood margin: 160.83686526923543

Actual digit: 6
Predicted digit: 1
Log-likelihood scores: [1921.25065118 1591.32116798 -2281.73199989]
Likelihood margin: 329.9294832033638

Actual digit: 6
Predicted digit: 9
Log-likelihood scores: [-1.98189836e+04 1.74900539e+01 8.85359225e+02]
Likelihood margin: 867.8691708518865

Actual digit: 9
Predicted digit: 1
Log-likelihood scores: [1183.41345796 -11641.3073663   1179.11149056]
Likelihood margin: 4.301967394652593

Actual digit: 9
Predicted digit: 6
Log-likelihood scores: [676.90675177 1346.66399312 1276.50025358]
Likelihood margin: 70.16373954111873

Conclusion:

Gaussian Mixture Models were implemented and evaluated for both synthetic two-dimensional data and handwritten digit images. In Part 1, separate three-component GMMs were trained for two classes using an 80:20 train-test split. The diagonal covariance model achieved the highest accuracy of 94.90%, compared with 94.75% for spherical covariance and 91.10% for full covariance. In Part 2, diagonal GMMs were trained for MNIST digits 1, 6 and 9. The models achieved 97.45% overall test accuracy. The confusion matrix, representative errors and likelihood margins were used to examine classification mistakes and the overlap between class models.